

1 Three Dimensional Coordinate Systems and Vectors

1.4 The Cross Product

Definition 1.4.1 (The Cross Product) If $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$, then the cross product, denoted $\vec{a} \times \vec{b}$ is given by

$$\vec{a} \times \vec{b} = \langle a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1 \rangle.$$

Example 1.4.2 Find the cross product between the vectors $\vec{a} = \langle 1, 3, 4 \rangle$ and $\vec{b} = \langle 2, 7, -5 \rangle$.

Theorem 1.4.3 *The vector $\vec{a} \times \vec{b}$ is orthogonal to both $\vec{a}$ and $\vec{b}$.*

Theorem 1.4.4 *If θ is the angle between vectors $\vec{a}$ and $\vec{b}$, then*

$$|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin \theta.$$

Theorem 1.4.5 *Two vectors $\vec{a}$ and $\vec{b}$ are parallel if and only if $\vec{a} \times \vec{b} = 0$.*

Example 1.4.6 Find a vector which is perpendicular to the plane passing through the points $P(1, 4, 6)$, $Q(-2, 5, -1)$, and $R(1, -1, 1)$.

Theorem 1.4.7 (*Properties of the Cross Product*) If $\vec{a}$, $\vec{b}$, and $\vec{c}$ are vectors and c is a scalar, then

1. $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$
2. $\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$
3. $(c\vec{a}) \times \vec{b} = c(\vec{a} \times \vec{b}) = \vec{a} \times (c\vec{b})$
4. $(\vec{a} + \vec{b}) \times \vec{c} = \vec{a} \times \vec{c} + \vec{b} \times \vec{c}$
5. $\vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \cdot \vec{c}$
6. $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$

The product $\vec{a} \cdot (\vec{b} \times \vec{c})$ that occurs in the above theorem is known as the scalar triple product. And its value is the volume of the parallelepiped determined by the vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$.

Example 1.4.8 Find the volume of the parallelepiped determined by the three vectors $\vec{a} = \langle 1, 4, -7 \rangle$, $\vec{b} = \langle 2, -1, 4 \rangle$, and $\vec{c} = \langle 0, -9, 18 \rangle$.

If we have a force $\vec{F}$ acting on a rigid body at a point given a position vector $\vec{r}$, the torque is defined $\vec{\tau}$ is defined to be the cross product of $\vec{r}$ and $\vec{F}$:

$$\vec{\tau} = \vec{r} \times \vec{F}.$$

The torque is a measure of the tendency of the body to rotate, and the direction of the torque vector is the axis of rotation (rotating clockwise about this axis)

Example 1.4.9 A bolt is tightened by applying a 40 N force to a 0.25 m wrench. Find the magnitude of the torque about the center of the bolt.